MILITARY SPECIFICATION

PROJECTILE, 155MM, HE, RA, M549A1 FORWARD (M37 EXTRUDED) PROPELLANT GRAIN FOR

This specification is approved for use by the US Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification contains requirements not covered by the drawings and provides quality assurance provisions for the fabrication of the rocket motor forward propellant grain for the 155MM Rocket Assisted Projectile, M549A1.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-F-320 - Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes
PPP-T-60 - Tape, Packing, Water Proof
PPP-T-76 - Tape, Pressure Sensitive Adhesive Paper, Water Resistant (for carton Sealing)
PPP-B-636 - Box, Shipping, Fiberboard

MILITARY

MIL-B-117 - Bag, Sleeve and Tubing - Interior Packaging
MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

FSC 1320

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn. DRSMC-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-P-50578 - M549 Projectile, Loading, Assembly and Packing
MIL-P-63201 - Propellant Solventless Type M37 for Projectile 155MM, HE, Rocket, M549
MIL-B-2427 - Box, Ammunition, Packing, Wood, Nailed
MIL-F-50449 - Filler, Sheet Form

STANDARDS

MILITARY

MIL-STD-453 - Inspection, Radiographic
MIL-STD-105 - Inspection by Attributes

2.1.1 Other government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS (see 6.6)

U.S. ARMY ARMAMENT MUNITIONS AND CHEMICAL COMMAND (AMCCOM)

PRODUCT AND PACKAGING DRAWINGS*

9324787 - Grain Assy, Forward
9324782-1 - Propellant, Billet
8796522 - Marking Diagram and Sealing for Wood Packing Boxes

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS Title 49, Transportation, Parts 100-199.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C., 20402. Orders for the above should cite: 49 CFR 100-199 (latest revision)).

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with applicable drawings and specifications.

3.1.1 Use of Class 1 propellant. The Carpet Roll shall conform to the requirements of Military Specification MIL-P-63201.

3.1.2 Use of Class 2 propellant. The Billet shall conform to the requirements of Military Specification MIL-P-63201.

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3.2 Propellant grain. The inhibited propellant grain shall comply with the requirements specified on drawing 9324787 and secondary drawings and with all requirements specified on applicable specifications.

3.3 Cracks, foreign material and cavitation

3.3.1 Radiographic examination

a. Cracks - There shall be no cracks as shown on the radiograph as dark lines in excess of .06 inches in length.

b. Foreign material - There shall be no foreign material in excess of 0.06 inches in any shape or dimension as shown by bright spots on the radiograph.

c. Cavitation - There shall be no cavitation as shown by low density on the radiograph in excess of .03 square inches individually or a maximum combined area of .06 square inches in any 3 inches of length.

3.3.2 Visual examination

a. Cracks - There shall be no cracks present within the propellant.

b. Foreign material - There shall be no foreign material in excess of 0.06 inches in any shape or dimension.

c. Cavitation - There shall be no cavitation in excess of .03 square inches individually or a maximum combined area of .06 square inches in any 3 inches of length.

3.4 Inhibitors

3.4.1 Separation and delamination. The propellant grain shall contain no individual separation between the inhibitor and the grain or between the inhibitor layers (delamination) which is greater than 0.010 square inch. The total combined area of propellant inhibitor separations and delaminations shall be no greater than 0.020 square inch. There shall be no propellant inhibitor separation within one inch of any split surface.

3.4.2 Exposed propellant. There shall be no exposed propellant at the bond lines of inhibitor segments.

3.5 Physical properties. Compression load and deformation.

3.5.1 Plus 140°F. The average of three (3) maximum compression readings from each lot shall be no less than 465 lbs., and the average of three (3) deformation readings from each lot shall be no less than 69 percent when tested as specified in 4.4.3.3.1.

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3.5.2 Minus 40°F. The average of three (3) compression readings from each lot shall be no less than 9877 pounds and the average of three (3) deformation readings from each lot shall be no less than 11 percent when tested as specified in 4.4.3.3.2.

3.6 First article inspection. This specification makes provisions for both preproduction and initial production testing for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.7 Workmanship

3.7.1 Propellant, grain assembly. The propellant grain assembly shall be free of dirt, grease, chips, gouges, solvent drippings, and other defects which would make the item unsuitable for its intended use.

3.7.2 Inhibitors. The inhibitors (circumferential and disc) shall be free from dirt, grease, chips, gouges, cracks, and other defects which would make the item unsuitable for its intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First article inspection
b. Quality conformance inspection

4.3 First article inspection

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.1.2. The first article sample shall consist of the following items in sample quantities as indicated:

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Part Description	Drawing	Quantity
Grain Assy Fwd	9324787	50
Propellant Fwd Grain	9324706	13
Propellant Fwd grain (prior to splitting)	9324706	10
Propellant Billet (wrapped)	9324783-1	10
Inhibitor End Disc	9324784-1	10
Propellant, Billet	9324782-1	10

4.3.1.1 Process changes. Any proposed change in the manufacturing process must be reviewed by the design agency, who at their discretion may waive or may increase the first article requirements for the Grain Assy, Fwd. The increased first article will require a quantity of Sixty Four (64) Grain Assy, Fwd. All other first article requirements will remain unchanged.

4.3.2 Inspections to be performed. The first article parts submitted in accordance with paragraph 4.3.1 will be subjected by the Government to any or all the requirements of the applicable drawings and specifications and to the inspections of Table I. All inspections listed in Table I shall be classified Major unless otherwise specified in referenced paragraphs and specifications. See MIL-A-48078.

4.3.2.1 Ballistic tests.

4.3.2.1.1 Submission. Fifty (50) complete propellant grains packed in the form that will be used for shipping of the regular production quantity shall be submitted to a Government Installation as designated by the Contracting Officer for loading and ballistic testing. Unless otherwise directed by the Contracting Officer, the propellant grains shall not be submitted prior to the acceptance of the sample submitted to the Government for examination (see 4.3.1).

4.3.2.1.2 Tests to be performed. Proving Ground Tests shall be performed for information only using the procedures of 4.5.4.

4.3.2.2 Process changes. In the event the increased first article in 4.3.1.1 is required, the Sixty Four (64) Grain Assy, Fwd shall be inspected as follows:

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Examination or Test	No. of Samples	Requir Para.	Para. Ref/ Inspection Method
Examination for Defects and Tests	64	3.2	4.4.2.6
Radiographic Inspection	64	3.3.1	4.5.1.2
12.2 Meter Drop Test (-40°F)	12	Info	TOP-4-2-601
12.2 Meter Drop Test (-60°F)	12	Info	TOP-4-2-601
Ballistic Test (-40°F)	10	3.2	4.5.4
Ballistic Test (-65°F)	10	3.2	4.5.4
Ballistic Test (+70°F)	10	3.2	4.5.4
Ballistic Test (+145°F)	10	3.2	4.5.4

The ballistic and drop test samples shall be loaded into inert warhead/live rocket motor M549A1 projectiles. After the drop test is completed, the sample projectiles shall be radiographed in two orientations 90° apart. The radiographs as well as other test results will be submitted to ARRADCOM, DRDAR-QAR-Q to determine if a ballistic test is required.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE: Propellant Grain Fwd		SHEET 1 OF 1		DRAWING NUMBER: See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Inhibitor, End Disc (Dwg. 9324784-1)				
	Examination for Defects and Tests	10	-	3.2	4.4.2.1
	Propellant Billet (Wrapped) (Dwg. 9324783-1)				
	Examination for Defects and Tests	10	-	3.2	4.4.2.3
	Propellant, Fwd Grain (Priorto Splitting) (Dwg. 9324706)				
	Examination for Defects and Tests	10	-	3.2	4.4.2.4
	Propellant; Fwd Grain (Dwg. 9324706)				
	Examination for Defects and Tests	10	-	3.2	4.4.2.5
	Physical Properties	3	-	3.5	4.5.5
	Grain Assy, Fwd (Dwg. 9324787)				
	Examination for Defects and Tests	50	-	3.2	4.4.2.6
	Ballistic Test	50	-	3.2	4.5.4
	Radiographic Inspection	50	-	3.3.1	4.5.1.2
	Propellant, Billet (Dwg. 9324782-1)				
	Examination for Defects and Tests	10	-	3.2	4.4.2.2

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4.4 Quality conformance inspection

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

In addition, each lot of propellant grains shall contain not more than one lot or propellant. The lot interfix number shall be changed when a new lot of propellant is introduced into the rocket grain extruding process. Additionally, a grain lot shall be no less than one week or more than one month scheduled production of rocket grains.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test tables, sampling plans for Major and Minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9324784
4.4.2.1	Inhibitor, End Disc				NEXT HIGHER ASSEMBLY 9324706
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Thickness	a	0.40%	3.2	Gage
102	Excessive warpage	a	0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship	a	1.0%	3.7.2	Visual

NOTES:

a. See Paragraph 4.4.3.1.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET		DRAWING NUMBER
4.4.2.2	Propellant, Billet (After OD Machining)		1 OF 1		9324782-1
					NEXT HIGHER ASSEMBLY 9324783-1
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Outside diameter		0.40%	3.2	Gage
Minor 201	Evidence of poor workmanship		100%	3.7.1	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9324783-1
4.4.2.3	Propellant Billet (Wrapped)				NEXT HIGHER ASSEMBLY 9324706
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter		0.40%	3.2	Gage
102	Inhibitor improperly wrapped		0.40%	3.2	Visual
103	Radiographic inspection		100%	3.3.1	4.5.1.1
Minor					
201	Evidence of poor workmanship		1.0%	3.7.1	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.4	Propellant, Forward Grain (Prior to Splitting)				9324706
					NEXT HIGHER ASSEMBLY 9324787
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Total length (max.)		0.40%	3.2	Gage
102	True position of I.D. to O.D.		0.40%	3.2	Gage
103	Bevel missing		0.40%	3.2	Visual
Minor					
201	Inside diameter		0.65%	3.2	Gage
202	Evidence of poor workmanship		1.0%	3.7.1	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9324706
4.4.2.5	Propellant, Forward Grain				NEXT HIGHER ASSEMBLY 9324787
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Excessive cracks, foreign material or cavitation		100%	3.3.2	Visual
102	Exposed propellant		100%	3.4.2	Visual
103	Inhibitor separation or delamination		0.40%	3.4.1	Visual
104	Surface condition of saw cut surface		0.40%	3.2	Visual
105	Finished weight of grain		0.40%	3.2	Balance
106	Width of sawcut	a	-	3.2	Test/4.5.2
107	True position of sawcut	a	-	3.2	Test/4.5.3
Minor					
201	Diameter of pellet holes		0.65%	3.2	Gage
202	Depth of pellet holes		0.65%	3.2	Gage
203	True position of pellet holes		0.65%	3.2	Gage (See Para. 6.6)
204	Evidence of poor workmanship		1.0%	3.7.1	Visual

NOTES:
a. The first item at the beginning of each 8 hours of production shall be used for this test.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9324787
4.4.2.6	Grain Assembly, Forward				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Forward end improperly marked		0.40%	3.2	Visual
102	Spacers missing or improper		0.40%	3.2	Visual
103	Excessive adhesives		0.40%	3.2	Visual
104	Pellet above flush, missing or damaged		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		1.0%	3.7.1	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.7	Packing, Forward Grain				See Para. 5.1 & 5.2
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Protective bag damaged		0.40%	5.1	Visual
102	Tape missing or improperly applied		0.40%	5.1	Visual
103	Drum split or damaged		0.40%	5.1	Visual
Minor					
201	Marking incomplete or incorrect		0.65%	5.2	Visual
202	Evidence of poor workmanship		1.0%	3.7.1	Visual

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4.4.3 Testing

4.4.3.1 Inhibitor end disc inspections. The following sampling sequence may be used in lieu of sampling per the AQL's listed in paragraph 4.4.2.1.

4.4.3.1.1 The first consecutively produced pieces from each set of dies or mold cavities shall be inspected for dimensional characteristics as shown on the applicable drawing and applicable paragraph in this specification until twenty (20) consecutively produced acceptable items are found. If five hundred (500) pieces are produced without obtaining 20 consecutive acceptable items, production shall be stopped, and the process corrected prior to accepting any additional components.

4.4.3.1.2 After meeting the requirements of 4.4.3.1.1, five (5) pieces from each set of dies shall be selected at intervals of 1000 pieces per set of dies or mold cavities and inspected for dimensional characteristics as specified on the applicable drawing and applicable paragraph in this specification. The occurrence of any defect on any sample piece shall be cause for taking corrective action, for rejecting all pieces produced since the last acceptable sample, and returning to sampling per 4.4.3.1.1.

4.4.3.2 Radiographic inspection.

4.4.3.2.1 Billet inspection - Major defect. The propellant billet shall be subjected to 100% radiographic inspection. This test shall be conducted in accordance with 4.5.1.1.

4.4.3.2.2 Grain inspection - Major defect. One sample of a completed grain selected per 8 hours production shall be radiographed. In the event the sample grain fails to meet the requirements of 3.3.1, all grains produced after the previous radiographic sample shall be radiographed and any additional grains failing to meet the requirements of 3.3.1 shall be removed from the lot. This test shall be conducted in accordance with 4.5.1.2.

4.4.3.3 Physical properties.

4.4.3.3.1 Plus 140°F - Major defect. Three (3) samples will be randomly selected from the lot. If possible, no two samples will be taken from the same billet. These samples shall be tested for maximum compression load and deformation as specified in 4.5.5. The lot shall be rejected if the average of the three specimens from any casting fails to meet the requirements of 3.5.1.

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4.4.3.3.2 Minus 40°F - Major defect. Three (3) samples will be randomly selected for the lot. If possible, no two samples will be taken from the same billet. These samples shall be tested for compression load and deformation as specified in 4.5.5. The lot shall be rejected if the average of the three specimens from any casting fails to meet the requirements of 3.5.2.

4.4.3.4 Width of cut - Major defect. The width of cut taken in splitting the grain shall not exceed the drawing requirement. For verification, the width of cut shall be inspected at the beginning of each 8 hours of production for compliance with 3.2.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the 'Paragraph Reference/Inspection Method' column in the tables starting with Paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and Section 6.3 herein.

4.5 Test methods and procedures.

4.5.1 Radiographic inspection.

4.5.1.1 Billet inspection. The propellant billet shall be radiographed with the beam parallel to the longitudinal axis of the billet. Observations shall be made for cracks, voids, fissures and foreign material in accordance with MIL-STD-453. A penetrameter of sample test block showing the types of defects for rejection shall be designed to assure a 2 percent minimum sensitivity.

4.5.1.1.1 Fluoroscope. In lieu of x-ray equipment, fluoroscopic equipment can be utilized for radiographic inspection of the propellant billet. The fluoroscopic equipment shall have the capability to permanently record the fluoroscopic images, record the comments of the inspector, photograph selected defects, and measure defect sizes. An Image Quality Indicator should be in the field of view at all times to simulate cavity and crack type defects.

4.5.1.2 Grain inspection. The split finished machined propellant grain shall be radiographed with the beam parallel to the longitudinal axis of the grain. Observations shall be made for cracks, voids, fissures and foreign material in accordance with 3.3.1. Radiographic inspection procedures shall be in accordance

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with MIL-STD-453. A penetrameter or sample test block showing the types of defects for rejection shall be designed to assure a 2 percent minimum sensitivity.

4.5.2 Width of saw cut. The width of the sawcut shall be measured by making a partial cut through the sample or paragraph 4.4.2.6 and then removing the saw blade and measuring the width of cut.

4.5.3 True position of saw cut. The sample used in 4.5.2 shall be gaged after the partial cut for true position of the saw cuts.

4.5.4 Ballistic test. The 50 grain assemblies manufactured during the first article shall be inspected per Table I and then shipped to a specified Government loading facility. These grains shall be loaded into a M549 (Inert Warhead/Live motor Body) test vehicle and fired at a Government Test Facility. Test procedures to be used are those contained in the M549 Projectile Loading, Assembly and Packing specification (MIL-P-50578). Failure of the propelling grains to perform according to the criteria of MIL-P-50578 shall be cause for rejection of the first article sample.

4.5.5 Physical properties. Each specimen and specimen support shall conform to the requirements of Figure 1. The compression testing shall be conducted in a test fixture equipped with a temperature conditioning box as an integral part of the fixture. The specimen shall be inserted between the fixture posts with steel cylinders the same diameter as the specimen and placed above and below the specimen forming one continuous concentric cylinder. The specimen shall be conditioned at the required temperature plus or minus 5°F for three hours minimum and shall be tested at a crosshead speed of 2 inches per minute.

5. Packaging

5.1 Specific requirements

5.1.1 Level A. Twelve (12) forward grains shall be packaged in a RSC Style, B-Flute, SW, Domestic Class, 175 pound (minimum), fiberboard box in accordance with specification PPP-B-636. The inside size of the fiberbox is to be 17 x 11 3/8 x 8 1/4 inches (+ 1/8 tolerance). Place spacer (11 x 5 1/2 inches) between flaps (top and bottom).

The grains shall be packaged in two layers of six (6) per layer. Each grain shall be inserted into a spiral or convolute wound chipboard tube having the following dimensions: Inside

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diameter 5.290 - .015; Outside diameter 5.620 - .030, Length 4.000 - .020 inches. A separator (16 1/2 x 11.0 inches) of the same material as the fiber box shall be placed between layers. Additional separators shall be placed on the top, and on the sides (16 1/2 x 8. inches) and the ends (10 3/4 x 8. inches) as required for tight packing. The maximum moisture content for paper products shall be 9% prior to packout.

Close fiber box and seal with a center sealing strip using Class 1, Type III, any color, Spec PPP-T-60 tape. Blunt corners of fiber box and insert into a Type I, Class E, Style 1, Specification MIL-B-117, Barrier bag. The size of the barrier bag shall be approximately 21 1/4 inches open face by 29.0 inches deep. Exhaust excess air within the bag and seal it in accordance with the manufacturer's instructions.

5.1.2 Level B - See 5.2.2.

5.2. Packing

5.2.1 Level A. Over pack a bagged fiber box in a grade A, Type II, Class 1 Specification MIL-B-2427 wood box. The inside diameter of the box shall be 18 1/8 x 11 7/8 x 9.0 inches (+ 1/8 tolerance). One filler shall be used on the inside of the box, one at each end and one on the top. Additional fillers shall be used at the end, side and top as needed for tight packing. The fillers shall comply with spec MIL-F-50449. The fillers shall not be less than 1/2 inch smaller than the inside diameter of the box. The box cover shall be put in place and the box shall be strapped with two straps in accordance with MIL-B-2427.

5.2.2 Level B. (Not For Overseas Shipments) - Twelve (12) forward grains shall be packed in a fiber drum meeting the requirements of DOT specifications 21C of the Code of Federal Regulations Title 49 Parts 100 to 199. The drum wall shall be lined with a kraft backed commercial grade single faced corrugated fiber board using as many layers as required, for a tight pack. The drum size shall be approximately 11 5/8 by 11 5/8 inches by 13 1/4 inches high. A barrier bag meeting the requirements of MIL-B-117, Type I, Class E, Style 1 shall be placed inside the lined drum and twelve (12) propellant grains shall be packed in 3 layers of 4 per layer. The barrier bag size shall be approximately 28 3/4 inches long by 22 1/4 inches wide. Each grain shall be inserted into a spiral or convolute wound chipboard tube having the following dimensions: Inside diameter 5.290 + .015; outside diameter 5.620 - .030; length 4.000 - .020 inches. The tube winding shall be secured with dextrin or polyvinyl adhesive and the inside wall surface shall be smooth.

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Each grain shall be placed in the fiber drum with marked end up. A separator shall be placed between layers and on top of the final layer. The separator shall meet the requirements of PPP-F-320, Type CF, Class Domestic, Grade 200, B or C Flute material or equivalent. The bag shall be folded and sealed with tape meeting the requirements of PPP-T-60. Filler constructed of the same material as the separators shall be placed on top of the contents as required to provide a tight pack. The fiber drum cover shall be sealed with a 2-inch minimum wide tape meeting the requirements of PPP-T-76. All packing material inside the barrier bag shall not exceed 9 percent maximum moisture content.

5.3 Marking.

5.3.1 Level A. The fiber box and bag shall be marked with 1/4 inch (minimum) high lettering. Marking on the bag shall be with waterproof black ink. The lettering shall appear as follows:

"Grain Propelling Forward f/RAP 155MM, M549A1
Date Packed (Mo/Yr)
Lot No. (Insert No.)"

The wood box is to be marked in accordance with Drawing 8796522. DOT to be "Propellant Explosives (Solid) Class B" Nomenclature as above.

5.3.2 Level B. The fiber drum shall be marked as follows:

Marking on Top

"Propellant Explosives (Solid) Class B"

Marking on Side Wall

"12 Grain, Propellant, Forward f/RAP 155MM, M549A1
Date Packed (Mo./Yr.)
Lot No. (Insert No.)
Wt. (Insert Wt.) Cu (Insert Volume)"

Marking shall be with black waterproof ink and lettering shall be at least 1/4 inch high.

6. NOTES

6.1 Intended use. The components covered by this specification are intended for use on the 155MM Rocket Assisted Projectile, M549A1.

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6.2 Ordering data. See MIL-A-48078.

6.3 Submission of inspection procedures and/or equipment designs for approval- See MIL-A-48078. Submit equipment designs as required, to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAR-I, Dover, New Jersey 07801.

6.4 Inspection of true position of pellet holes (See 4.4.2.5). In lieu of performing this inspection the contractor may verify the true position by virtue of tooling by submitting appropriate designs for approval to control the true position.

6.5 Inspection of true position of saw cut - (See 4.4.2.5). In lieu of performing this inspection, the contractor may verify the true position by virtue of tooling by submitting appropriate designs for approval to control the true position.

6.6 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

Custodian:
Army - AR

Preparing activity:
Army - AR

(Project 1320-A434)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-63202	PROJECTILE, 155MM, HE, RA, M549A1 FORWARD (M37 extruded)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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